Shreya Vardhan

Contact Information

Institute for Quantum Information and Matter
California Institute of Technology
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Research Interests

Quantum many-body chaos, black hole physics, entanglement dynamics, and hydrodynamics.

Employment

California Institute of Technology May 2025-present
Postdoctoral Scholar, Institute for Quantum Information and Matter
Host: John Preskill

Stanford University August 2022-May 2025
Postdoctoral Researcher, Stanford Institute for Theoretical Physics
Hosts: Douglas Stanford, Stephen Shenker

Education

Massachusetts Institute of Technology 2017-2022
Ph.D. in Physics
Advisor: Hong Liu
Thesis title: Chaos and thermalization in quantum many-body systems and gravity

Harvard University 2013-2017
Bachelor of Arts, *Magna Cum Laude* with highest honors in Physics and Mathematics

Delhi Public School, R. K. Puram, New Delhi 2011-2013

Publications

Author ordering is alphabetical unless otherwise noted. * indicates equal contribution.

1. *An efficient classical algorithm for OTOCs in chaotic quantum dynamics*
   Yunchao Liu, Yihui Quek, Shreya Vardhan, and Tianci Zhou
   [2611.12773[quant-ph]]
2. *Negative energies and the breakdown of bulk geometry*
   John Preskill, Mykhaylo Usatyuk, and Shreya Vardhan
   Accepted to Physical Review D [2603.25782[hep-th]]
3. *Free mutual information and higher-point OTOCs*
   Shreya Vardhan and Jinzhao Wang
   PRX Quantum **7**, 043017 [2509.13406[quant-ph]]
4. *Chirality, magic, and quantum correlations in multipartite quantum states*
   Shreya Vardhan, Bowen Shi, Isaac H. Kim, and Yijian Zou (non-alphabetical)
   SciPost Physics 20, 066 (2026) [2503.10764 [quant-ph]]
5. *Estimating time in quantum chaotic systems and black holes*
   Haifeng Tang, Shreya Vardhan, and Jinzhao Wang
   SciPost Physics 19, 095 (2025) [2412.19885 [quant-ph]]

6. *Effective field theories of dissipative fluids with one-form symmetries*
   Shreya Vardhan, Saso Grozdanov, Samuel Leutheusser, and Hong Liu (non-alphabetical)
   Journal of High Energy Physics **2025**, 184 (2025) [2408.12868 [hep-th]]

7. *Entanglement dynamics from universal low-lying modes*
   Shreya Vardhan and Sanjay Moudgalya (non-alphabetical)
   Physical Review B **113**, 014308 [2407.16763 [cond-mat.stat-mech]]

8. *Scramblon loops*
   Douglas Stanford, Shreya Vardhan, and Shunyu Yao
   Journal of High Energy Physics **2024**, 73 (2024) [2311.12121[hep-th]]

9. *Petz recovery from subsystems in conformal field theory*
   Shreya Vardhan, Annie Wei, and Yijian Zou
   Journal of High Energy Physics **2024**, 16 (2024) [2307.14434 [hep-th]]

10. *Local dynamics and the structure of chaotic eigenstates*
    Zhengyan Darius Shi*, Shreya Vardhan*, and Hong Liu (non-alphabetical)
    Physical Review B **108**, 224305 [2306.08032 [quant-ph]]

11. *The black hole interior from non-isometric codes and complexity*
    Chris Akers, Netta Engelhardt, Daniel Harlow, Geoff Penington, and Shreya Vardhan
    Journal of High Energy Physics **2024**, 155 (2024) [arXiv:2207.06536 [hep-th]]

12. *Strong-field magnetohydrodynamics for neutron stars*
    Shreya Vardhan, Sašo Grozdanov, Samuel Leutheusser, and Hong Liu (non-alphabetical)
    Physical Review Research **6**, L042050 [arXiv:2207.01636 [astro-ph.HE]]

13. *Mixed-state entanglement and information recovery in thermalized states and evaporating black holes*
    Shreya Vardhan*, Jonah Kudler-Flam*, Hassan Shapourian, and Hong Liu (non-alphabetical)
    Journal of High Energy Physics **2023**, 64 (2023) [arXiv:2112.00020 [hep-th]]

14. *Bound entanglement in thermalized states and black hole radiation*
    Shreya Vardhan*, Jonah Kudler-Flam*, Hassan Shapourian, and Hong Liu (non-alphabetical)
    Physical Review Letters **129**, 061602 [arXiv:2110.02959 [hep-th]]

15. *Entanglement entropies of equilibrated pure states in quantum many-body systems and gravity*
    Hong Liu and Shreya Vardhan
    PRX Quantum **2**, 010344 [arXiv:2008.01089 [hep-th]]

16. *A dynamical mechanism for the Page curve from quantum chaos*
    Hong Liu and Shreya Vardhan
    Journal of High Energy Physics **2021**, 88 (2021) [arXiv:2002.05734 [hep-th]]

17. *Void formation in operator growth, entanglement, and unitarity*
    Hong Liu and Shreya Vardhan
    Journal of High Energy Physics **2021**, 159 (2021) [arXiv:1912.08918 [quant-ph]]

18. *Characterizing time-irreversibility in disordered fermionic systems by the effect of local perturbations*
Shreya Vardhan, Giuseppe De Tomasi, Markus Heyl, Eric J. Heller, and Frank Pollmann (non-alphabetical)
Physical Review Letters **119**, 016802 (2017) [arXiv:1702.01622 [cond-mat.dis-nn]]

Invited Conference Talks

July *Negative energies and the breakdown of bulk geometry* **2026**
Advances in Black Hole Theory, Indian Institute of Science Education and Research, Pune

June *Free mutual information and higher-point OTOCs*
Contours 2026, University of Cambridge

June *Fidelity estimation from quantum many-body chaos*
Quantum Chaotic Many-Body Dynamics 2026, University of Warwick

January *Free mutual information as a measure of quantum chaos*
Quantum Information Processing (QIP) Conference, University of Latvia

December *Free mutual information and higher-point OTOCs* **2025**
Hydrodynamics, Fluctuations, and Noise in Quantum and Classical Systems, ICTS, Bangalore

June *The free mutual information as a measure of quantum chaos*
Quantum Information in Quantum Gravity Workshop, Perimeter Institute, Waterloo, Canada

November *Estimating time in quantum chaotic systems and black holes* **2024**
"Strings and Particles in Ohio, Cincinnati, and Kentucky" (SPOCK) Symposium, University of Cincinnati

November *Estimating time in quantum chaotic systems*
"Bay Area Strings, Information and Cosmology Symposium" (BASICS), University of California, Berkeley

August *Entanglement dynamics from universal low-lying modes*
"Quantum Information, Quantum Field Theory, and Gravity," International Centre for Theoretical Sciences, Bangalore

June *Entanglement dynamics from universal low-lying modes*
Strings Conference, CERN, Geneva

April *Hydrodynamic pictures for entanglement growth*
Spacetime and String Theory Conference, Kavli Institute for Theoretical Physics, UC Santa Barbara

February *Lessons for quantum gravity from quantum chaos*
"What is String Theory? Weaving Perspectives Together" workshop at Kavli Institute for Theoretical Physics, UC Santa Barbara

September *Local dynamics and the structure of chaotic eigenstates* **2023**
Long Term Workshop on Quantum Information, Quantum Matter and Quantum Gravity, Yukawa Institute for Theoretical Physics, Kyoto University

July *Petz recovery from subsystems in conformal field theory*
It From Qubit Conference, Perimeter Institute, Waterloo, Canada

Invited Seminars

October *Chaos, complexity, and the breakdown of semiclassical gravity* **2026**
Theory Colloquium, University of Cologne

October *Negative energies and the breakdown of bulk geometry*
Qubits and Spacetime Seminar, Okinawa Institute of Science and Technology

July *Free mutual information and higher-point OTOCs*
Mathematical Physics Seminar, University of Oxford

April *Negative energies and the breakdown of bulk geometry*
Graduate Theory Seminar, Yale University

March *Negative energies and the breakdown of bulk lengths in JT gravity*
High Energy Physics Seminar, Chennai Mathematical Institute

March *Negative energies and the breakdown of bulk lengths in JT gravity*
Holography Seminar, University of Texas, Austin

February *Negative energies and the breakdown of bulk lengths in JT gravity*
High Energy Theory Seminar, University of Michigan

November *Free mutual information and higher-point OTOCs* **2025**
Many-body Physics Seminar, Eisert Group, Freie Universitaet Berlin

June *Quantum chaotic systems and black holes: puzzles and lessons from an information-theoretic perspective*
Physics Department Seminar, University of British Columbia

February *Dynamics of quantum entanglement in quantum many-body systems and gravity*
Department Colloquium, University of Kentucky

November *Estimating time in quantum chaotic systems and black holes* **2024**
High Energy Theory Seminar, Purdue University

November *Estimating time in quantum chaotic systems and black holes*
Black Hole Initiative Colloquium, Harvard University

November *Entanglement dynamics from universal low-lying modes*
String Seminar, University of Kentucky

November *Estimating time in quantum chaotic systems and black holes*
Theoretical Elementary Particle Seminar, University of California, Los Angeles

October *Entanglement dynamics from universal low-lying modes*
Mathematical and Theoretical Physics Seminar, University of Illinois at Urbana-Champaign

August *Entanglement dynamics from universal low-lying modes*
Quantum Spacetime Seminar Series, Tata Institute for Fundamental Research, Mumbai

June *Entanglement dynamics from universal low-lying modes*
Condensed Matter Seminar, Technische University, Munich

May *Emergence of the membrane picture in Brownian models*
Mezei Group Meeting, Oxford University

March *Emergence and breakdown of the membrane picture in Brownian models*
String Seminar, University of California, Berkeley

March *Emergence and breakdown of the membrane picture in Brownian models*
QFT/Holography Zoom Seminar, University of Bristol

February *Emergence and breakdown of the membrane picture in Brownian models*
String Seminar, California Institute of Technology

February *Petz recovery from subsystems in conformal field theory*
String Seminar, University of California, San Diego

October *Local dynamics and the structure of chaotic eigenstates* **2023**
Statistical Physics Seminar, International Center for Theoretical Sciences, Bangalore

October *Petz recovery from subsystems in conformal field theory*
High Energy Theory Seminar, International Center for Theoretical Sciences, Bangalore

August *Petz recovery from subsystems in conformal field theory*
High Energy Theory Seminar, IIT Madras

May *Non-isometric codes and the black hole interior*
The Quantum Algorithms and Applications Collaboratory (QuAAC) seminar series, Sandia National Laboratory

March *Non-isometric codes and the black hole interior*
High Energy Theory Seminar, University of Michigan

January-February *Understanding recent progress on the black hole information loss paradox: series of 4 lectures*
Lecture 1: Introduction to the black hole information problem
Lecture 2: The equilibrium approximation
Lecture 3: Non-isometric codes
Lecture 4: Complexity and the black hole interior
Tata-Infosys Lecture Series, Tata Institute for Fundamental Research, Mumbai

November *Entanglement dynamics at finite temperature* **2022**
Center for Quantum Mathematics and Physics (QMAP) Seminar, University of California, Davis

September *Non-isometric codes and the black hole interior*
High Energy Theory Seminar, CERN

April *Mixed-state entanglement and information recovery in thermalized states and evaporating black holes*
Berkeley Center for Theoretical Physics Seminar, University of California, Berkeley

March *Mixed-state entanglement and information recovery in evaporating black holes*
Theoretical High-Energy Physics Seminar, McGill University

February *The entanglement structure of evaporating black holes at finite temperature*
Holography Journal Club, University of Texas at Austin

February *Mixed-state entanglement and information recovery in evaporating black holes*
String Seminar, International Center for Theoretical Sciences, Bangalore

December *Mixed-state entanglement and information recovery in thermalized states and evaporating black holes* **2021**
High Energy Theory Seminar, Perimeter Institute, Waterloo

October *Bound entanglement in thermalized states and black hole radiation*
Quantum Spacetime Seminar Series, Tata Institute of Fundamental Research (TIFR), Mumbai

May *The entanglement structure of equilibrated pure states*
University of Kentucky High Energy Theory Seminar

March *Entanglement entropies of equilibrated pure states and the origin of replica wormholes*
University of Ljubljana Holography Group Meeting

March *Entanglement entropies of equilibrated pure states and the origin of replica wormholes*
University of Pennsylvania High Energy Theory Seminar

October *A universal approximation for entanglement entropies of equilibrated pure states* **2020**
Quantum Chaos Online Seminar Series

October *A universal approximation for entanglement entropies of equilibrated pure states*
Harvard/MIT Joint Condensed Matter Seminar

May *Void formation, unitarity, and growth of multipartite entanglement*
Quantum Fluids in Isolation Virtual Seminar Series, Boston College

April *Void formation in operator growth, entanglement, and unitarity*
Physics Department Lunch Talk, Massachusetts Institute of Technology

Teaching

- *Physics of Solids* Teaching Assistant, MIT — Fall 2021
- *Graduate Quantum Mechanics I* Teaching Assistant, MIT — Fall 2020
- *Theory of Solids II* Teaching Assistant, MIT — Spring 2020
- *Physics III: Oscillations and Waves* Teaching Assistant, MIT — Spring 2019
- *Graduate Quantum Mechanics I* Teaching Assistant, MIT — Fall 2018
- *Electrodynamics* Homework grader, Harvard University — Fall 2015

Service and Outreach

- Referee for Journal of High Energy Physics (12), Physical Review Letters (2), IOP Reports on Progress in Physics (1), Physical Review D (1), PRX Quantum (1), and Quantum Information Processing Conference 2026 and 2027 (2) — 2022-present
- Volunteered for the “theory lab tour” at the “FUTURE of Physics” program for undergraduate students held at Caltech — September 2025
- Organizer of Q-Farm Workshop on “Thermalization and Quantum Information,” Stanford — October 2024
- Gave a talk on “Quantum chaos and the black hole information paradox” at Quantum High School Lecture Series, Palo Alto — May 2024
- SITP Seminar Organizer, Stanford — 2022-present

- MIT Physics Colloquium committee representative and student lunch organizer 2018-2021
- Mentor for Physics Department Directed Reading Program, MIT January 2020

Honors and Awards

- Election to Harvard Chapter of the Phi Beta Kappa academic honor society. May 2017
- Harvard College Program for Research in Science and Engineering (PRISE) Fellow. May 2015
- Jacob Wendell Scholarship Prize for the Harvard College sophomore identified by the selection committee as the most academically promising and broad-ranging scholar in his or her class. February 2015
- Detur Book Prize for outstanding academic performance in the first year at Harvard College. October 2014
- Times Scholarship, national level academic honor in India. 2012

Programming languages

Proficient in Python, Mathematica, and C++
Familiar with MATLAB and SQL

Languages

Fluent in English and Hindi
Intermediate Russian